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Emerging SEM equipment system combat capability assessment method

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Abstract

The system combat capability reflects the general characteristics of the weapon system. The combat capability of the equipment system is the result of the interaction and interaction of each system, and the emergence of the system has not yet formed a clear research idea and measurement method for its combat capability. Aiming at this problem, this paper analyzes the emergent characteristics of the system's combat c The system combat capability reflects the general characteristics of the weapon system. The combat capability of the equipment system is the result of the interaction and interaction of each system, and the emergence of the system has not yet formed a clear research idea and measurement method for its combat capability. Aiming at this problem, this paper analyzes the emergent characteristics of the system's combat capability and the structure and relationship between the weapon systems, and constructs an evaluation model for the emergence of the combat capability of the equipment system based on the structural equation model. It can reflect the generation characteristics of the system's combat capability and the emerging sources, and provide a scientific basis for improving the combat effectiveness of the system, and also provide support for the evaluation of the contribution of the research equipment system. Taking an air defense and anti-missile warfare as an example, and the model parameters estimation and the example is evaluated and analyzed by using LISREL8.70 software. The effectiveness of the system's combat capability assessment model based on structural equation model is verified. Capability and the structure and relationship between the weapon systems, and constructs an evaluation model for the emergence of the combat capability of the equipment system based on the structural equation model. It can reflect the generation characteristics of the system's combat capability and the emerging sources, and provide a scientific basis for improving the combat effectiveness of the system, and also provide support for the evaluation of the contribution of the research equipment system. Taking an air defense and anti-missile warfare as an example, and the model parameters estimation and the example is evaluated and analyzed by using LISREL8.70 software. The effectiveness of the system's combat capability assessment model based on structural equation model is verified.

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1. Introduction

With the development of information technology, high-tech (artificial intelligence) will have a broad application prospect in the military field. As the future war is intelligent war, overall program and construction of intelligent equipment system supported by military technology and application is crucial to winning the war. Emergence is one of the basic characteristics of the system, which refers to the components of the system interact with each other to produce extra-new properties in "1+1 > 2"[1]. Describing and measuring the "emergence" of combat capability of equipment systems with traditional methods is very difficult.[2] Therefore, analyzing the emerging characteristics of these complex equipment systems is one of the core issues in the research of combat capability of equipment system.

To solve the above problems, this paper introduces a multivariate statistical method-Structural Equation Model (SEM) to analyze the combat capability of a system. By judging the correlation between the equipment system and subsystems and the mechanism, the combat capability of the equipment system can be evaluated.

2. SEM based combat capability model construction

Based on the aforementioned index and technical framework of equipment system combat capability assessment and the basic principles of SEM, the technical framework of equipment system combat capability assessment should be transformed into a mathematical language familiar with the structural equation. SEM has two kinds of variables: explicit variables and latent variables. Explicit variables are obtained by measurement, which are used for analysis and calculation by SEM, and latent variables are estimated by explicit variables. In addition, the variables affected by other variables are endogenous variables, while exogenous variables are not. Therefore, the variables of the structural equation model are actually divided into endogenous latent variables, exogenous latent variables, endogenous explicit variables and exogenous explicit variables. The structural equation model is formed by connecting latent variable and explicit variables through mathematical equations. The main object of our research is the emergence of equipment system, which is characterized by its combat capability, while the emergence of system combat capability is generated by a certain structural and functional relationship of system capability composition. In the process of capability index aggregation, the system can determine latent variables and structural equation models of variables, and show the validity of equipment system variables. Performance level capability index and latent variables are formed by the secondary capability index and equipment system capabilities, and the relationship between display variables and latent variables is determined by the parameters. The details are as shown in Figure 1 below. In the figure, the observed variables are represented by rectangles, and the latent variables are represented by ellipses.

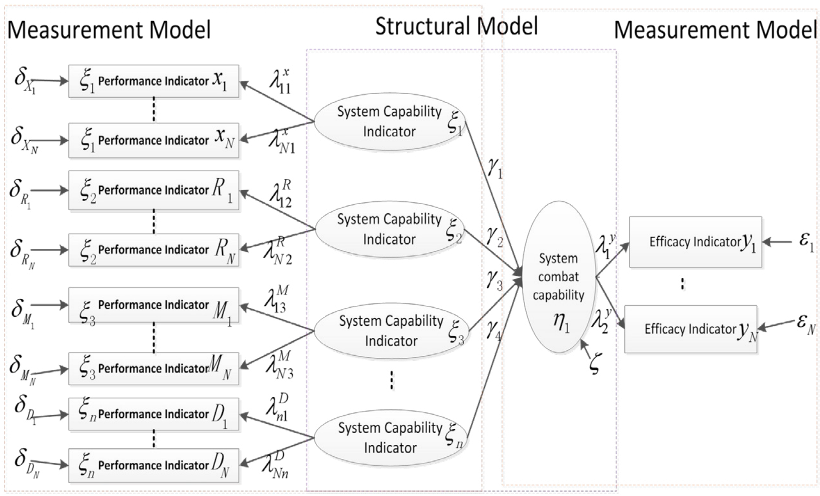

Fig. 1. SEM model for equipment combat capability assessment

The measurement equation of the model is:

$$X=\begin{pmatrix} x_1\\ \vdots\\ x_N\\ R_1\\ \vdots\\ R_N\\ M_1\\ \vdots\\ M_N\\ \vdots\\ D_1\\ \vdots\\ D_N \end{pmatrix}=\Lambda_x\xi+\delta=\begin{pmatrix} \lambda_{11}^x & 0 & 0 & 0\\ \vdots & & & \\ \lambda_{2N}^x & 0 & 0 & 0\\ 0 & \lambda_{12}^R & 0 & 0\\ & \vdots & & \\ 0 & \lambda_{N2}^R & 0 & 0\\ 0 & 0 & \lambda_{13}^M & 0\\ & & \vdots & \\ 0 & 0 & \lambda_{N3}^M & 0\\ & & \ddots & \\ 0 & 0 & 0 & \lambda_{1n}^D\\ & & & \vdots\\ 0 & 0 & 0 & \lambda_{Nn}^D \end{pmatrix}\begin{pmatrix} \xi_1\\ \xi_2\\ \xi_3\\ \vdots\\ \xi_n \end{pmatrix}+\begin{pmatrix} \delta_{x_1}\\ \vdots\\ \delta_{x_N}\\ \delta_{R_1}\\ \vdots\\ \delta_{R_N}\\ \delta_{M_1}\\ \vdots\\ \delta_{M_N}\\ \vdots\\ \delta_{D_1}\\ \vdots\\ \delta_{D_N} \end{pmatrix} \tag{1}$$

$$Y=\begin{pmatrix} y_1\\ \vdots\\ y_N \end{pmatrix}=\Lambda_y\eta+\varepsilon=\begin{pmatrix} \lambda_1^y\\ \vdots\\ \lambda_N^y \end{pmatrix}\eta+\begin{pmatrix} \varepsilon_1\\ \vdots\\ \varepsilon_N \end{pmatrix} \tag{2}$$

Where λ_{11}^x , λ_{2N}^x , λ_{12}^R , λ_{N2}^R , λ_{13}^M , λ_{N3}^M , λ_{1n}^D , λ_{Nn}^D , λ_1^y , λ_N^y are the path parameters of the correlation between the corresponding latent variables (capacity of each node layer) and the observation variables (capability index of performance layer); δ_{x_1} , δ_{x_N} , δ_{R_1} , δ_{R_N} , δ_{M_1} , δ_{M_N} , δ_{D_1} , δ_{D_N} , ε_1 , ε_N are the unexplained part of the observation variables (capability index of performance layer), namely the error term added to the corresponding observation variables.

The structural equation of the model is:

$$\eta=\gamma_1\xi_1+\gamma_2\xi_2+\gamma_3\xi_3+\cdots+\gamma_n\xi_n+\zeta \tag{3}$$

Where γ_1 , γ_2 , γ_3 , γ_4 is the path parameter of the relationship between the latent variable (the capability of each node layer ξ_1 ξ_2 ξ_3) and the latent variable η (the combat capability of the system). ζ is the part of the latent variable η (the combat capability of the system) that is not explained by the latent variable (the capability of each node layer ξ_1 ξ_2 ξ_3 ξ_n).

In this paper, the structural equation model of system equipment combat capability evaluation is established, and the problem of system combat capability evaluation is transformed into the problem of structural equation model parameter estimation of system combat capability, which can generate the influence degree of different equipment (systems) and their performance on weapon equipment system combat capability from mathematical logic. The structural equation model and measurement equation model are used to analyze the result of solving the index of the combat capability of the equipment system and the capability of each system, and the evaluation conclusion is given to determine the optimal composition scheme of the combat capability, so as to provide decision support for relevant departments.

3. Examples

In this paper, taking the air defense and anti-missile system of sea formation as an example, the effectiveness of the system operational capability evaluation method is verified.

As the combat of air defense and anti-missile system of maritime formation is a very complicated process, there are too many influential factors, which cannot be generalized. According to the theory of structural equation model, it is necessary to delimit the boundary of system research and select the key elements of the object of study for analysis.

3.1. SEM for combat effectiveness evaluation of air defense and anti-missile systems

According to the research on the definition, characteristics, structure, basic deployment and operational process of sea formation anti-aircraft missile system in related references [3-7], and response to the relationship between the operational process characteristics, hierarchical structure and index system framework of sea defense and anti-missile system, the combat capability evaluation technology of air defense missile system is brought into the structural equation framework, and the mathematical language is familiar, that is, the structural equation model of combat capability evaluation of sea formation anti-aircraft missile system is established, and the efficiency value of simulation results is input into LISREL 8.0 structural equation model. In Figure 2, E1 stands for reconnaissance and detection combat capability, E2 stands for command and decision combat capability, E3 stands for survival combat capability, and E4 stands for strike and intercept combat capability.

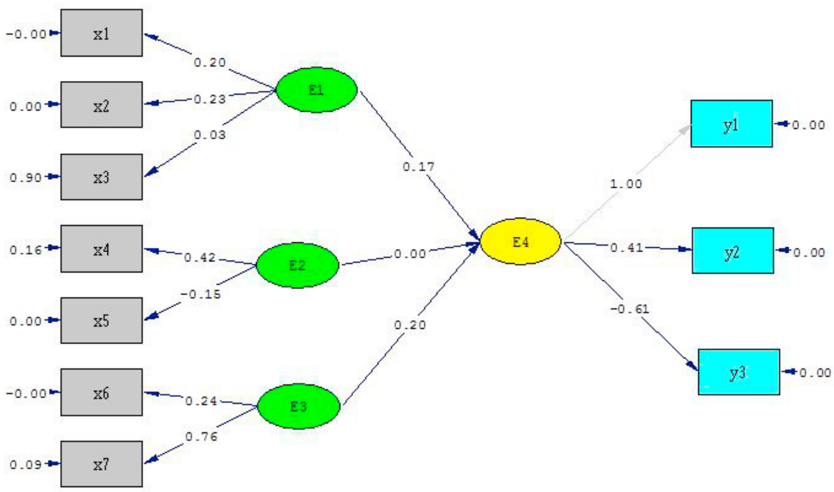

Fig. 2. SEM for combat capability assessment of air defense and anti-missile systems

3.2. Conclusion of emergence assessment of air defense and anti-missile system

By substituting the parameter values estimated by LISERL8.0 into the SEM model, the quantitative relationship model between the combat capability and combat effectiveness of the maritime formation in air defense and anti-missile combat is obtained as follows.

Evaluation model of reconnaissance and detection combat capability is

$$\xi_1 = 1.67x_1 + 1.45x_2 + 11.1x_3 - 10$$

Command and control capability assessment model is

$$\xi_2 = 1.19x_4 - 3.33x_5 - 0.19$$

Survivability assessment model is

$$\xi_3 = 2.08x_6 + 0.66x_7 - 0.06$$

Strike interception capability assessment model is

$$\eta_1 = 0.33y_1 + 0.81y_2 + 0.55y_3$$

According to the above formula, the evaluation values of combat capability of each system of the air defense and anti-missile system under 10 schemes are calculated, as shown in Table 1.

Table 1. Combat capability values for each system

Index group	ξ_1	ξ_2	ξ_3	η_1
1	5.93	0.69	8.23	2.65
2	3.06	3.34	2.69	1.06
3	1.19	0.39	1.27	0.05
4	7.25	2.60	0.90	1.43
5	5.92	0.92	0.92	0.82
6	3.57	0.29	0.90	0.43
7	32.33	0.36	0.85	5.67
8	5.89	2.34	0.87	1.17
9	7.15	0.61	1.03	1.42
10	16.69	0.28	0.85	3.01

By analyzing Table 1, the following three conclusions can be drawn:

(1) In view of the air defense missile weapon equipment system, adopted in this paper, the structural equation model to evaluate combat capability, reconnaissance detection capability, command and control warfare capability and the fight against intercept capability negative, its main reason is that the sea fleet combat capability of air defense missile system is by spot detection system, command decision-making system and survival ability of collaborative interaction between system and detection system and the command decision - making system capacity in the process of polymerization, convergence happened between level, level of mutual influence, under the influence of nonlinear, makes them whole show negative emerging, It seriously affects the combat capability of the whole air defense and anti-missile system.

(2) In order to combat the interception capability, the combat capability evaluation value of Group 7, Group 10 and Group 1 are relatively high. Therefore, these three schemes are mainly considered when choosing the optimal scheme. The detection system of these three schemes have relatively high combat capability, which indicates that

detection system is an important means to improve the combat capability of air defense and anti-missile system against interception.

(3) From the above two points, it can be concluded that in order to have higher air defense and anti-missile combat capability, the sea formation detection system is very important, and the detection capability of the detection equipment needs to be strengthened. In addition, it is necessary to restrain some factors that lead to the negative emergence of the whole weapons and equipment system capabilities, so as to develop the combat capability of the whole weapon and equipment system in a favorable direction.

4. Conclusion

In this paper, the system capabilities evaluation of weapon and equipment is taken as the research object, the technical framework of equipment system capabilities index is analyzed, and the structural equation model of equipment system capabilities is constructed. The model can reflect the emergence source and measure the emergence degree of this part to a certain extent. However, the construction of system capability evaluation model for weapon and equipment is a very complicated problem. In the information age, more and more external factors and forms will be considered in the weapon and equipment system. This paper constructs a system combat capability evaluation model, and the emergence of this system can be said to be a useful attempt to apply the system combat capability evaluation method. Many shortcomings still need to be further studied. In the following research work, the emphasis will be placed on obtaining the sample data information of actual equipment indicators and optimizing the evaluation index system of the model, so as to improve the applicability and credibility of the model.

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